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Multi-Perspective Transformers in ARC-AGI-2 Challenge

A Hybrid Approach with Multi-View Representations and Custom Transformers

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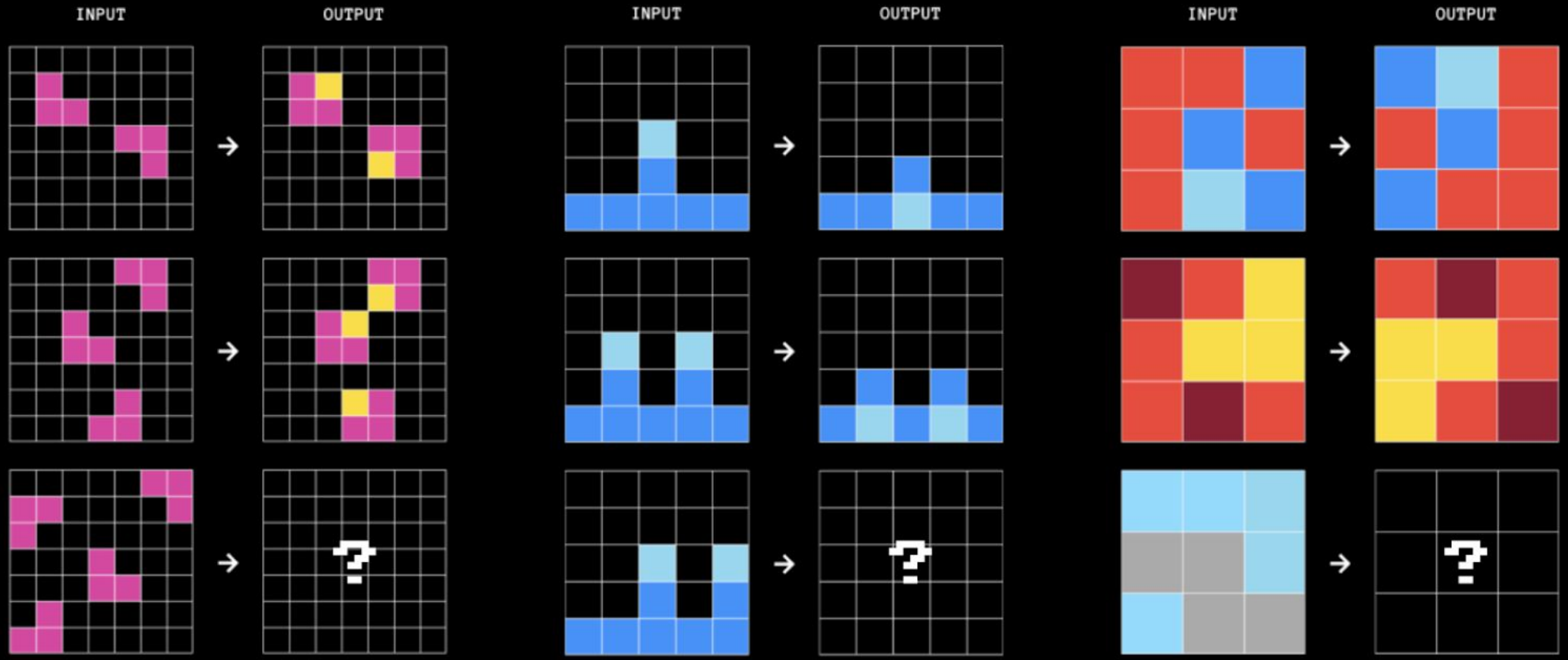
Motivation

What is ARC-AGI-2?

The Measure of Intelligence

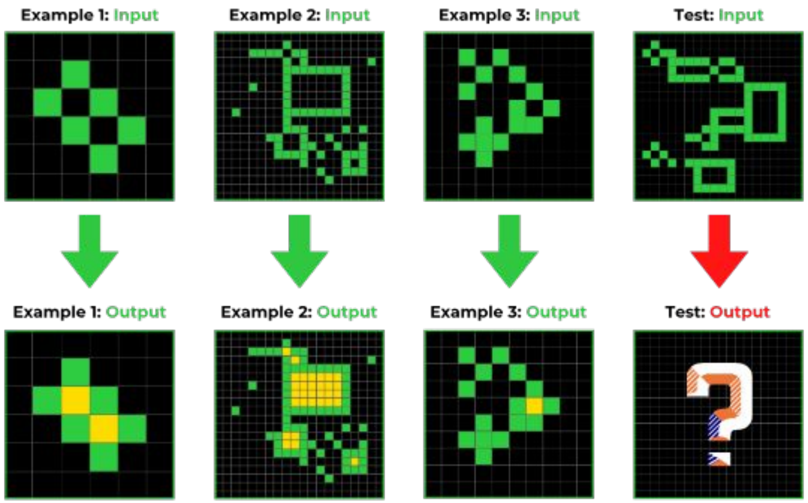
- Stands for **A**bstraction and **R**easoning **C**orpus for **A**rtificial **G**eneral **I**ntelligence
- Consists of human-intuitive tasks that challenges machines to
  - generalize from limited examples,
  - interpret symbolic meaning,
  - and flexibly apply rules in varying contexts



Example ARC-AGI tasks. Can you find the patterns?

Challenges

- **Few-shot learning**: a few input-output train pairs are provided
- **Abstract reasoning**
- **Visual** puzzles that are difficult to solve with brute force



Research Questions

The Core Problem

To design a model which is good at novel abstract logic

Requirements

A model that generalizes from a minimal number of examples to solve a novel, never-before-seen task with pixel-perfect accuracy and high efficiency

Why Existing Models Fail

Standard Transformers are data-hungry and struggle to interpret 2D spatial relationships when flattened into standard 1D sequences without specialized handling.

Vanilla Large Language Models (LLMs) excel at pattern matching from massive datasets but struggle with few-shot scenarios, without prior knowledge.

Convolutional Neural Networks (CNNs) capture local texture well but often miss the global logic required to infer the "rule" behind an ARC task.

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Methods

Input Augmentation

Learn the Rule, Not the Rendering

To train a robust model, we process every input grid with multiple possible augmentations:

- Geometric Transformations i.e. rotations, flips, transpose
- Color Permutations i.e. changing the colors (e.g., Red ↔ Blue) to prevent overfitting to specific hues.

The purpose: The model learns to solve the underlying pattern, regardless of orientation or color scheme.

Input Augmentation

- We do not store massive static datasets.
- In every epoch, we dynamically apply random geometry and color specifications.
- This grants scalability with variations of tasks and little memory overhead.

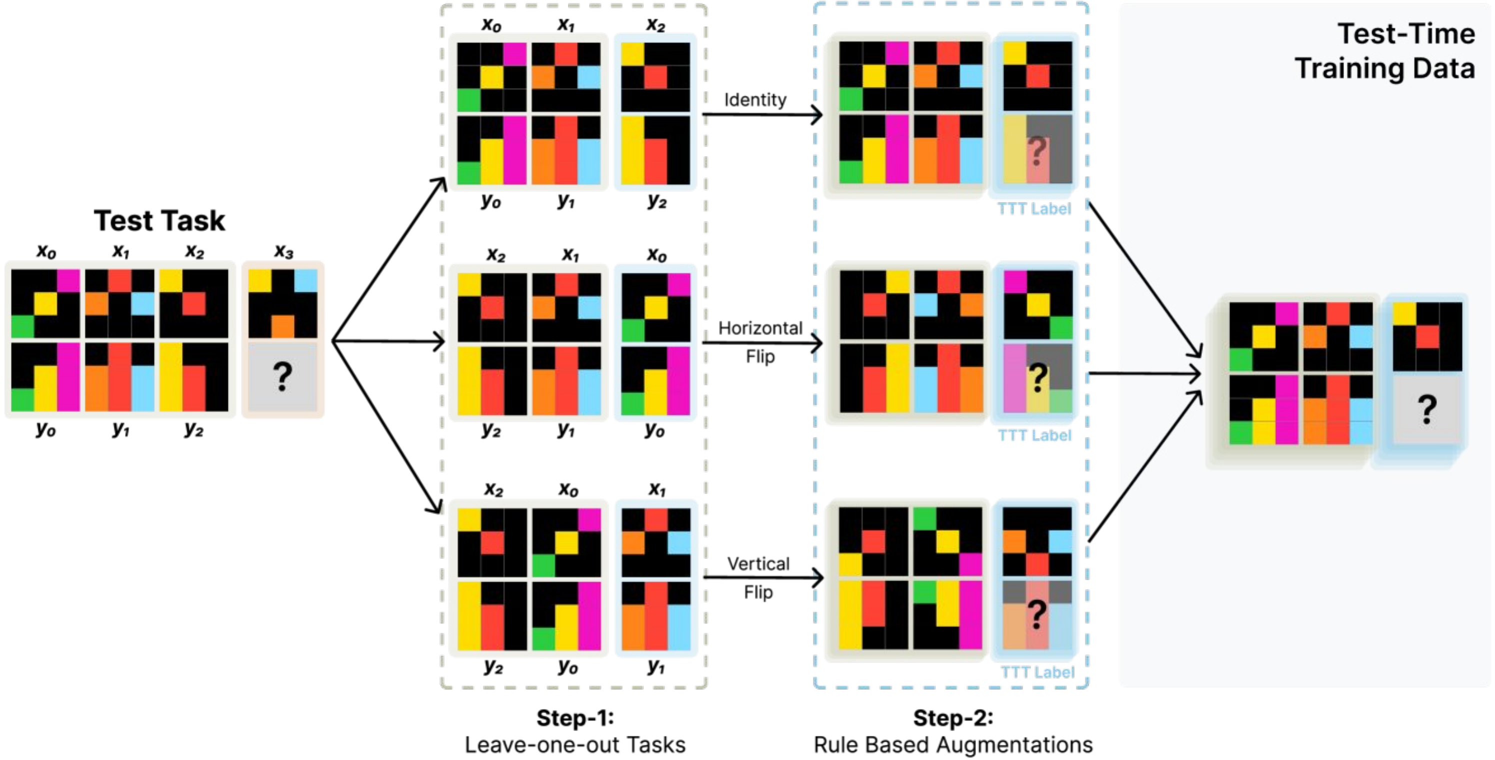
Core Architecture: TinyLM Model

- A Causal Self-Attention Transformer (GPT-style) tailored to abstract reasoning.
  - Depth: 8 Attention Blocks
  - Heads: 8 Parallel Attention Heads
  - Embedding Dimension: d model $\approx$ 448
  - Total Parameters: $\approx$ 20.3M

- Input Processing: Flattens 2D grids into 1D token sequences but utilizes Learned Positional Embeddings to strictly retain spatial awareness.

At inference time: Test-time Training (TTT)

Performs gradient updates on-the-fly based on test-time inputs



At inference time: Product of Experts (PoE)

- We generate predictions from multiple transformed "views" (rotated, flipped) of the same input
- Consensus:
    - Sum the log-probabilities from all views (filters out hallucinations).
    - The final answer is the one where multiple perspectives agree.

Findings

Results: Training Data

- Consists of 1000 Tasks

Models	**Qwen3-8B (Zero-shot)**	**Our Models**			
Metrics	Benchmark	Pre-trained Only	TTT	PoE	TTT + PoE
Accuracy	6.24%	**96.1%**	1.9%	23.8%	30.6%
Valid Generation Rate	99.4%	**96.1%**	1.9%	23.8%	30.6%
Failed Generation Rate	0.6%	**3.9%**	98.1%	76.2%	69.4%

Results: Test Data

- Consists of 120 Tasks

Models	**Qwen3-8B (Zero-shot)**	**Our Model**			
Metrics	Benchmark	Pre-trained Only	TTT	PoE	TTT + PoE
Accuracy	0%	**21.7%**	0%	0%	0%
Valid Generation Rate	96%	**21.7%**	0%	0%	0%
Failed Generation Rate	4%	**78.3%**	100%	100%	100%

Future work

- More testing with TTT and PoE
- Valid generation of predictions from the model
- Training more model with different parameters

Contributions

Caleb Talley

- Written: Project Proposal, Progress Report, Slides
- Code: Environment Setup, Repo Setup, Validate Dataset, Grid Representation + Operations, Test Cases, Model Scoring, Baseline Benchmarking
- General Logistics, Organization Setup, Method Brainstorming

Seun Adekunle

- Written: Project Proposal, Progress Report
- Code: Environment Setup, Repo Setup, View Grid, Evaluation, Test Cases, Code cleanup
- General Logistics, Organization Setup

Weiwen Dong

- Written: Project Proposal, Progress Report, Slides
- Code: Serialization, Baseline Benchmarking, Evaluation

Fariha Sheikh

- Built a simple runner with two guesses per task.
- Added time-budget and timing printouts.
- Made it offline and able to create a dummy submission.

Vedant Tibrewal

- Written:Project Proposal, Progress Report, Slides
- Code: Data augmentation, Model Training
- Method Brainstorming

Xinyu Wu

- Written: Project Proposal, Progress Report, Slides
- Code: Test-time training

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Questions